BS RESEARCH PROJECT

Forecasting the Inflation Trend in Pakistan Using a Vector Autoregressive (VAR) Model

Muhammad Arsim

Roll No: 2K20/STAT/91

Contents

1 Introduction

2 Literature Review

3 Objectives & Hypotheses

4 Data & Methodology

5 Results & Diagnostics

6 Forecasting

7 Conclusion

8 Recommendations & References

Introduction: What Is Inflation?

Inflation

The sustained increase in the general price level of goods and services over time, which reduces the purchasing power of a currency.

In Pakistan it is measured by the Consumer Price Index (CPI), and is influenced by oil prices and the exchange rate.





Inflation and Its Three Key Variables



CPI

Consumer Price Index: measures the average change in prices of goods and services.



Oil / Petrol Price

Rising petrol prices raise transportation costs and feed through to consumer prices.



Dollar Exchange Rate

A weaker rupee makes imports costlier, pushing consumer prices higher.

Literature Review

Author (Year)	Focus of Study
LeBlanc & Chinn (2004)	Influence of oil price changes on inflation in the US, France, UK, Japan and Germany.
Kiani (2008)	Impact of high oil prices on Pakistan's economy.
Aziz (2009)	Relationship between real interest rates, real oil prices and real exchange rates.
Shaari, Hussain & Abdullah (2012)	Effects of oil price shocks on inflation using VAR-VECM and Granger causality.
Saleem & Ahmad (2015)	Connection between a nation's inflation, currency value and exchange rates.
Iqbal & Razaq (2018)	Exchange rate (PKR per USD) and crude oil prices in Pakistan.

Research Objectives

01 Forecast the inflation trend in Pakistan.

02 Forecast the inflation rate for the next 6 months.

03 Discuss the influence of oil prices and the exchange rate on inflation.

04 Summarise findings: forecasts, solutions and recommendations.

Hypotheses

Hypothesis 1: Trend

H_o: There is no significant trend in the inflation rate in Pakistan.

H_1: There is a significant trend (upward or downward) in Pakistan's inflation rate over the study period (2018 to 2023).

Hypothesis 2: Oil & Exchange Rate

H_o: Changes in oil and exchange rate (dollar) prices do not significantly influence inflation in Pakistan.

H_1: Changes in oil and exchange rate (dollar) prices significantly influence inflation in Pakistan.

Data and Sources

Variables

CPI (inflation), exchange rate (PKR per USD) and oil price (Pak petroleum, PKR per litre).

Period

61 monthly observations, September 2018 to September 2023.

Sources

World Bank, IndexMundi, Federal Bureau of Statistics, State Bank of Pakistan (Monthly Bulletin and Annual Reports), TradingEconomics and Investing.com.

Methodology: VAR Model

1

Model

A Vector Autoregressive (VAR) model, introduced by Christopher Sims (1980), captures the joint dynamics of CPI, oil price and exchange rate.

2

Estimation

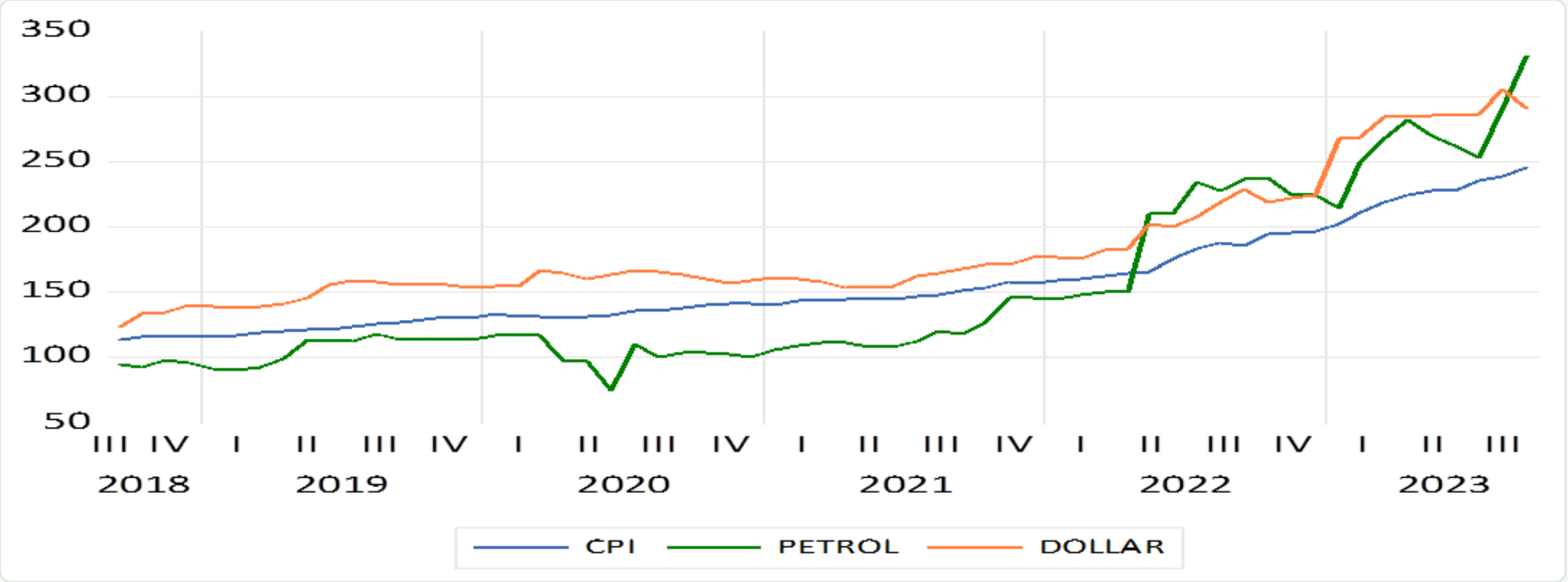
The model is estimated in EViews after testing stationarity, Granger causality and the optimal lag length.

3

Forecasting

The estimated VAR is used to forecast Pakistan’s inflation trend for the next 6 months.

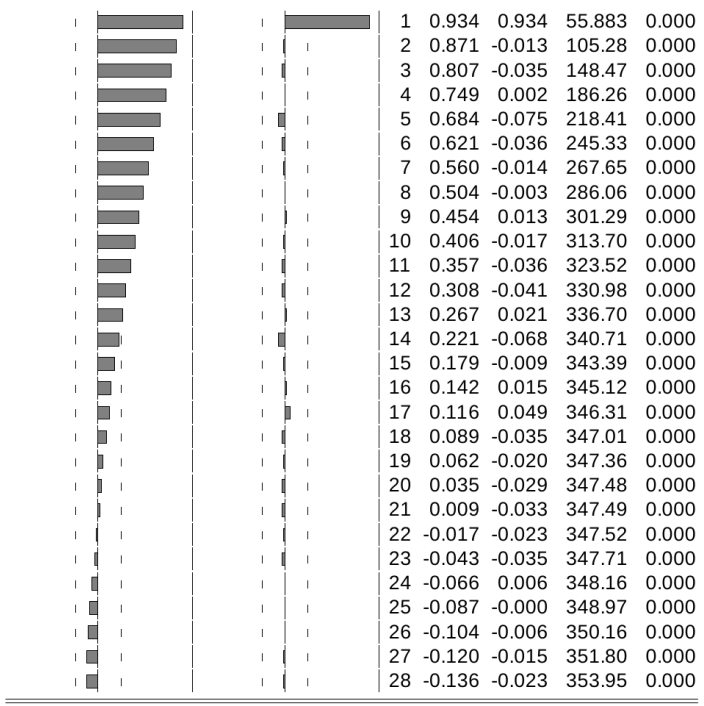
Monthly CPI, Petrol and Dollar Rate (Non-Stationary)



All three series show persistent upward movement and spikes, indicating non-stationarity. Differencing is applied before modelling.

Correlograms: Non-Stationary Series

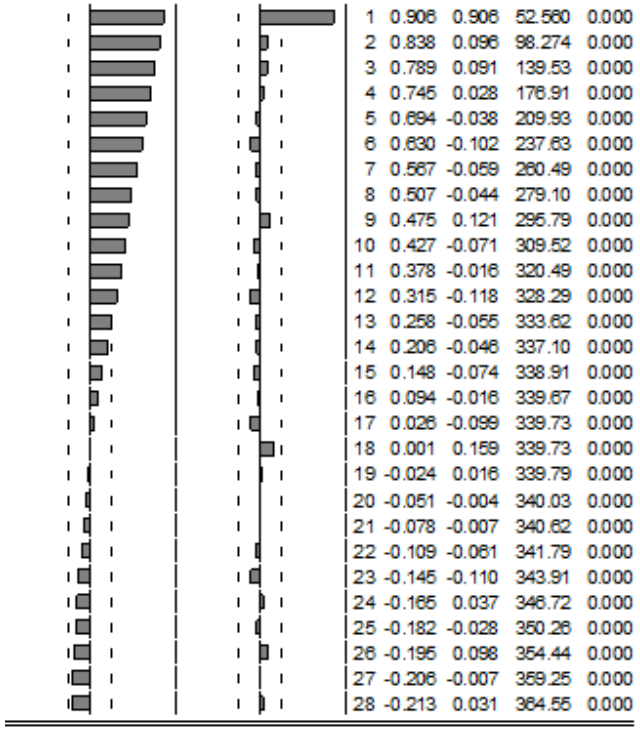
Date: 09/28/23 Time: 00:00
Sample: 2018M09 2023M09
Included observations: 61



	Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
1			0.934	0.934	55.883	0.000
2			0.871	-0.013	105.28	0.000
3			0.807	-0.035	148.47	0.000
4			0.749	0.002	186.26	0.000
5			0.684	-0.075	218.41	0.000
6			0.621	-0.036	245.33	0.000
7			0.560	-0.014	267.65	0.000
8			0.504	-0.003	286.06	0.000
9			0.454	0.013	301.29	0.000
10			0.406	-0.017	313.70	0.000
11			0.357	-0.036	323.52	0.000
12			0.308	-0.041	330.98	0.000
13			0.267	0.021	336.70	0.000
14			0.221	-0.068	340.71	0.000
15			0.179	-0.009	343.39	0.000
16			0.142	0.015	345.12	0.000
17			0.116	0.049	346.31	0.000
18			0.089	-0.035	347.01	0.000
19			0.062	-0.020	347.36	0.000
20			0.035	-0.029	347.48	0.000
21			0.009	-0.033	347.49	0.000
22			-0.017	-0.023	347.52	0.000
23			-0.043	-0.035	347.71	0.000
24			-0.066	0.006	348.16	0.000
25			-0.087	-0.000	348.97	0.000
26			-0.104	-0.006	350.16	0.000
27			-0.120	-0.015	351.80	0.000
28			-0.136	-0.023	353.95	0.000

CPI

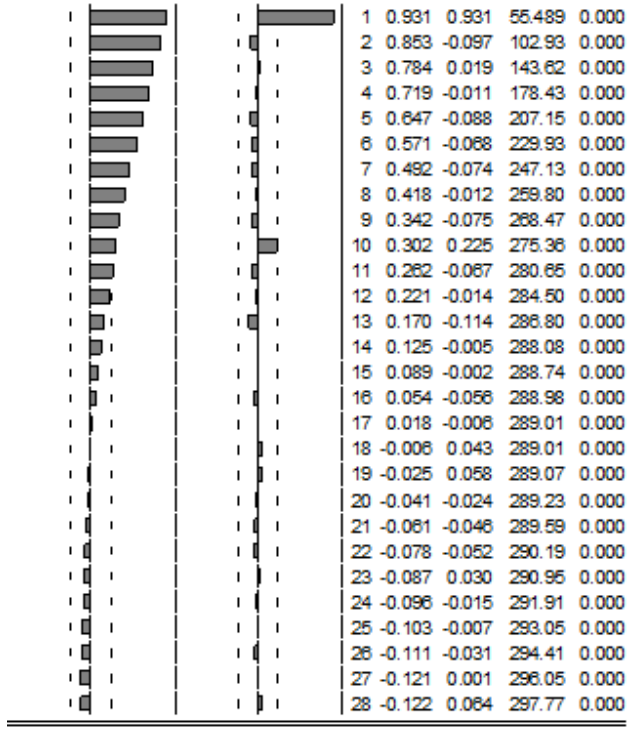
Date: 09/28/23 Time: 00:02
Sample: 2018M09 2023M09
Included observations: 61



	Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
1			0.906	0.906	52.560	0.000
2			0.838	0.096	98.274	0.000
3			0.789	0.091	139.53	0.000
4			0.745	0.028	176.91	0.000
5			0.694	-0.038	209.93	0.000
6			0.630	-0.102	237.63	0.000
7			0.567	-0.059	260.49	0.000
8			0.507	-0.044	279.10	0.000
9			0.475	0.121	295.79	0.000
10			0.427	-0.071	309.52	0.000
11			0.378	-0.016	320.49	0.000
12			0.315	-0.118	328.29	0.000
13			0.258	-0.055	333.62	0.000
14			0.206	-0.046	337.10	0.000
15			0.148	-0.074	338.91	0.000
16			0.094	-0.016	339.67	0.000
17			0.026	-0.099	339.73	0.000
18			0.001	0.159	339.73	0.000
19			-0.024	0.016	339.79	0.000
20			-0.051	-0.004	340.03	0.000
21			-0.078	-0.007	340.62	0.000
22			-0.109	-0.061	341.79	0.000
23			-0.145	-0.110	343.91	0.000
24			-0.165	0.037	346.72	0.000
25			-0.182	-0.028	350.26	0.000
26			-0.195	0.098	354.44	0.000
27			-0.206	-0.007	359.25	0.000
28			-0.213	0.031	364.55	0.000

Petrol

Date: 09/28/23 Time: 00:02
Sample: 2018M09 2023M09
Included observations: 61

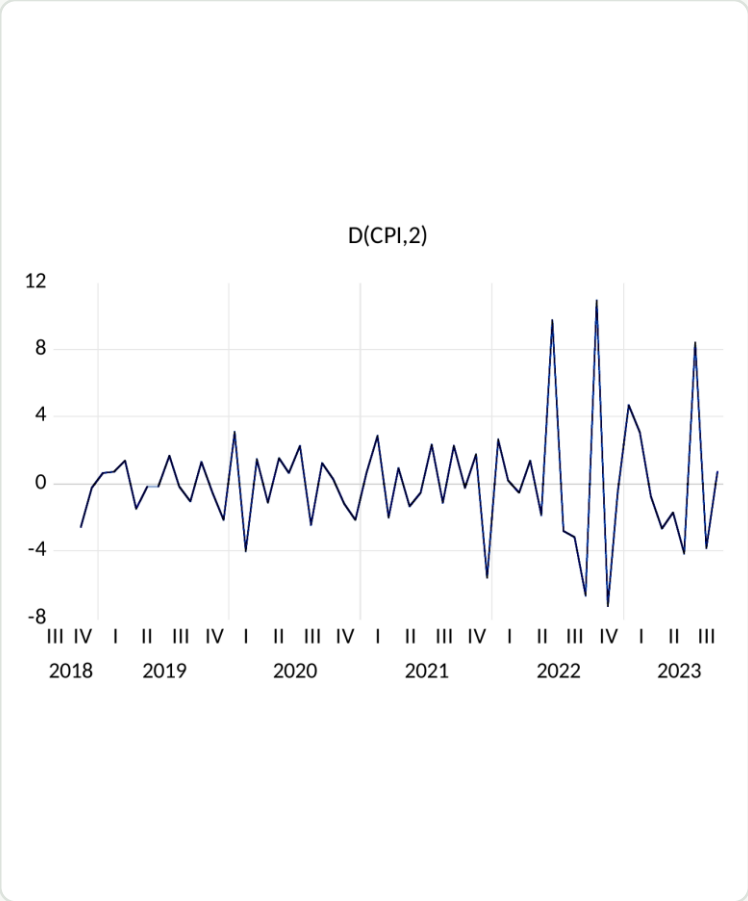


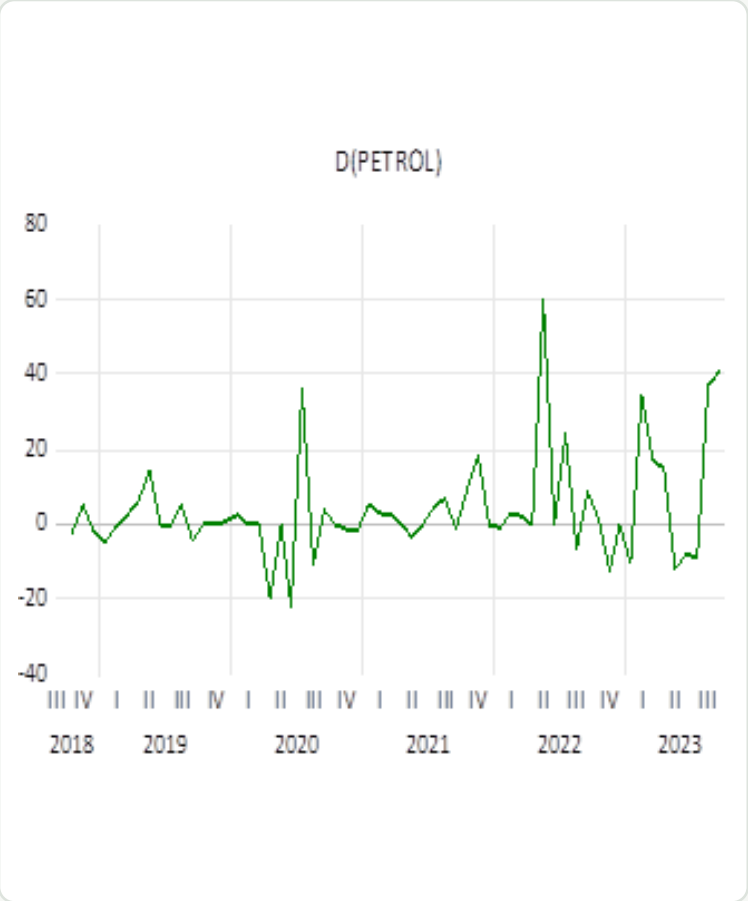
	Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
1			0.931	0.931	55.489	0.000
2			0.853	-0.097	102.93	0.000
3			0.784	0.019	143.62	0.000
4			0.719	-0.011	178.43	0.000
5			0.647	-0.088	207.15	0.000
6			0.571	-0.068	229.93	0.000
7			0.492	-0.074	247.13	0.000
8			0.418	-0.012	259.80	0.000
9			0.342	-0.075	268.47	0.000
10			0.302	0.225	275.36	0.000
11			0.262	-0.067	280.65	0.000
12			0.221	-0.014	284.50	0.000
13			0.170	-0.114	286.80	0.000
14			0.125	-0.005	288.08	0.000
15			0.089	-0.002	288.74	0.000
16			0.054	-0.056	288.98	0.000
17			0.018	-0.006	289.01	0.000
18			-0.006	0.043	289.01	0.000
19			-0.025	0.058	289.07	0.000
20			-0.041	-0.024	289.23	0.000
21			-0.061	-0.046	289.59	0.000
22			-0.078	-0.052	290.19	0.000
23			-0.087	0.030	290.95	0.000
24			-0.096	-0.015	291.91	0.000
25			-0.103	-0.007	293.05	0.000
26			-0.111	-0.031	294.41	0.000
27			-0.121	0.001	296.05	0.000
28			-0.122	0.064	297.77	0.000

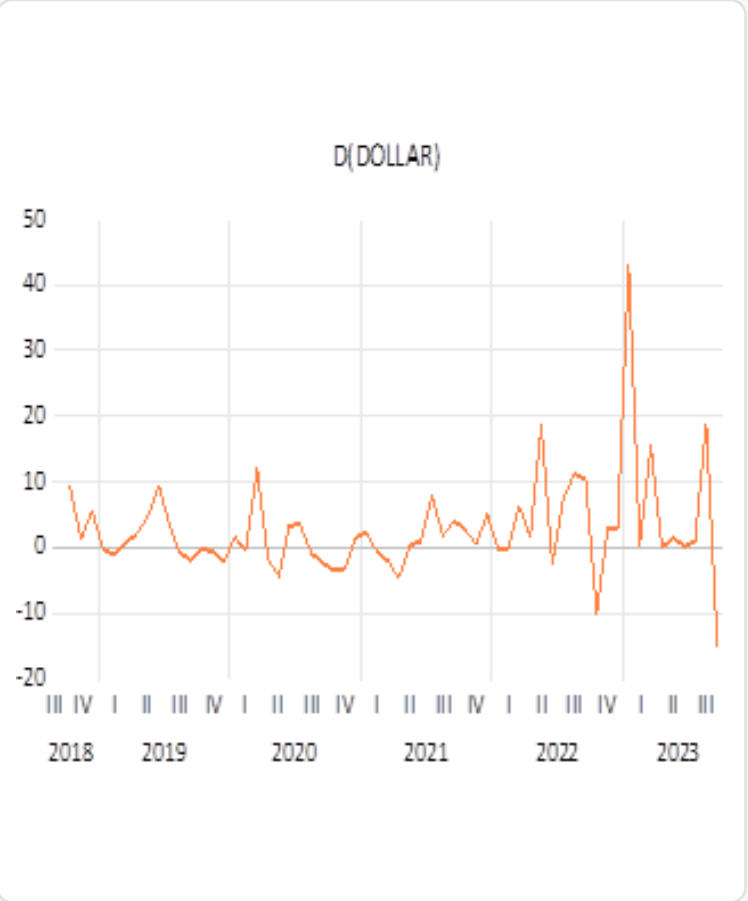
Dollar

Slowly decaying autocorrelations confirm non-stationarity in levels.

Stationary Series After Differencing





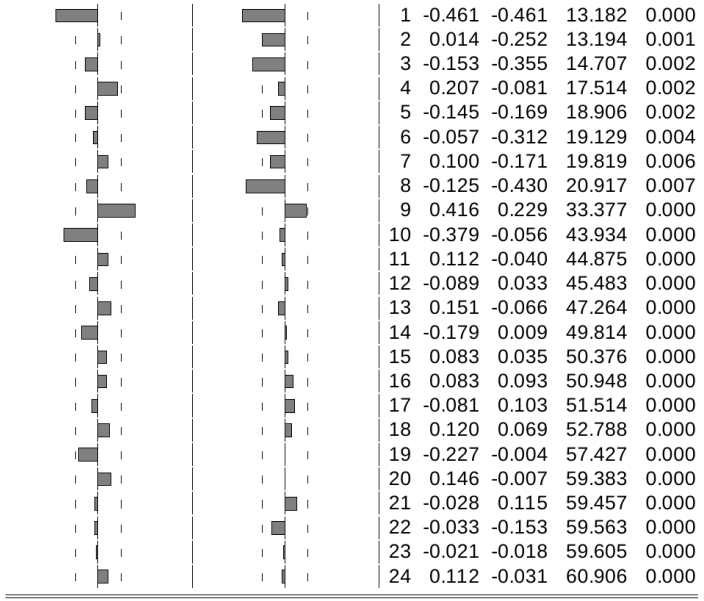


D(CPI, 2) **D(PETROL)** **D(DOLLAR)**

Fluctuations settle around a constant mean after differencing.

Correlograms: Stationary Series

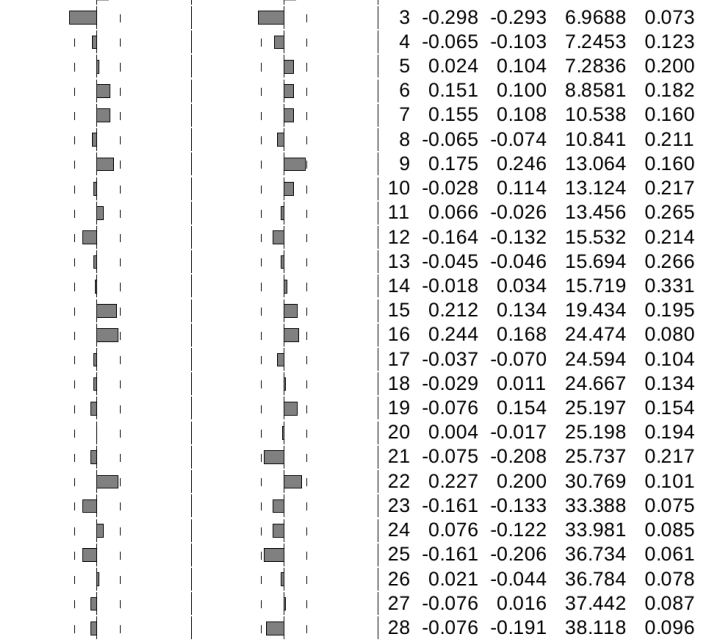
Date: 10/04/23 Time: 21:07
Sample (adjusted): 2018M11 2023M09
Included observations: 59 after adjustments



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	-0.461	-0.461	13.182	0.000
		2	0.014	-0.252	13.194	0.001
		3	-0.153	-0.355	14.707	0.002
		4	0.207	-0.081	17.514	0.002
		5	-0.145	-0.169	18.906	0.002
		6	-0.057	-0.312	19.129	0.004
		7	0.100	-0.171	19.819	0.006
		8	-0.125	-0.430	20.917	0.007
		9	0.416	0.229	33.377	0.000
		10	-0.379	-0.056	43.934	0.000
		11	0.112	-0.040	44.875	0.000
		12	-0.089	0.033	45.483	0.000
		13	0.151	-0.066	47.264	0.000
		14	-0.179	0.009	49.814	0.000
		15	0.083	0.035	50.376	0.000
		16	0.083	0.093	50.948	0.000
		17	-0.081	0.103	51.514	0.000
		18	0.120	0.069	52.788	0.000
		19	-0.227	-0.004	57.427	0.000
		20	0.146	-0.007	59.383	0.000
		21	-0.028	0.115	59.457	0.000
		22	-0.033	-0.153	59.563	0.000
		23	-0.021	-0.018	59.605	0.000
		24	0.112	-0.031	60.906	0.000

D(CPI, 2)

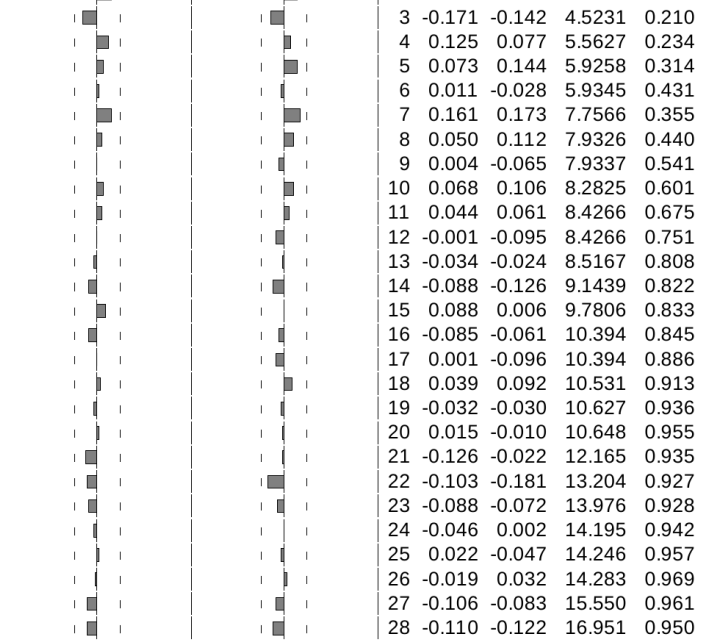
Date: 10/04/23 Time: 21:08
Sample (adjusted): 2018M10 2023M09
Included observations: 60 after adjustments



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	-0.044	-0.044	0.1235	0.725
		2	0.127	0.126	1.1642	0.559
		3	-0.298	-0.293	6.9688	0.073
		4	-0.065	-0.103	7.2453	0.123
		5	0.024	0.104	7.2836	0.200
		6	0.151	0.100	8.8581	0.182
		7	0.155	0.108	10.538	0.160
		8	-0.065	-0.074	10.841	0.211
		9	0.175	0.246	13.064	0.160
		10	-0.028	0.114	13.124	0.217
		11	0.066	-0.026	13.456	0.265
		12	-0.164	-0.132	15.532	0.214
		13	-0.045	-0.046	15.694	0.266
		14	-0.018	0.034	15.719	0.331
		15	0.212	0.134	19.434	0.195
		16	0.244	0.168	24.474	0.080
		17	-0.037	-0.070	24.594	0.104
		18	-0.029	0.011	24.667	0.134
		19	-0.076	0.154	25.197	0.154
		20	0.004	-0.017	25.198	0.194
		21	-0.075	-0.208	25.737	0.217
		22	0.227	0.200	30.769	0.101
		23	-0.161	-0.133	33.388	0.075
		24	0.076	-0.122	33.981	0.085
		25	-0.161	-0.206	36.734	0.061
		26	0.021	-0.044	36.784	0.078
		27	-0.076	0.016	37.442	0.087
		28	-0.076	-0.191	38.118	0.096

D(PETROL)

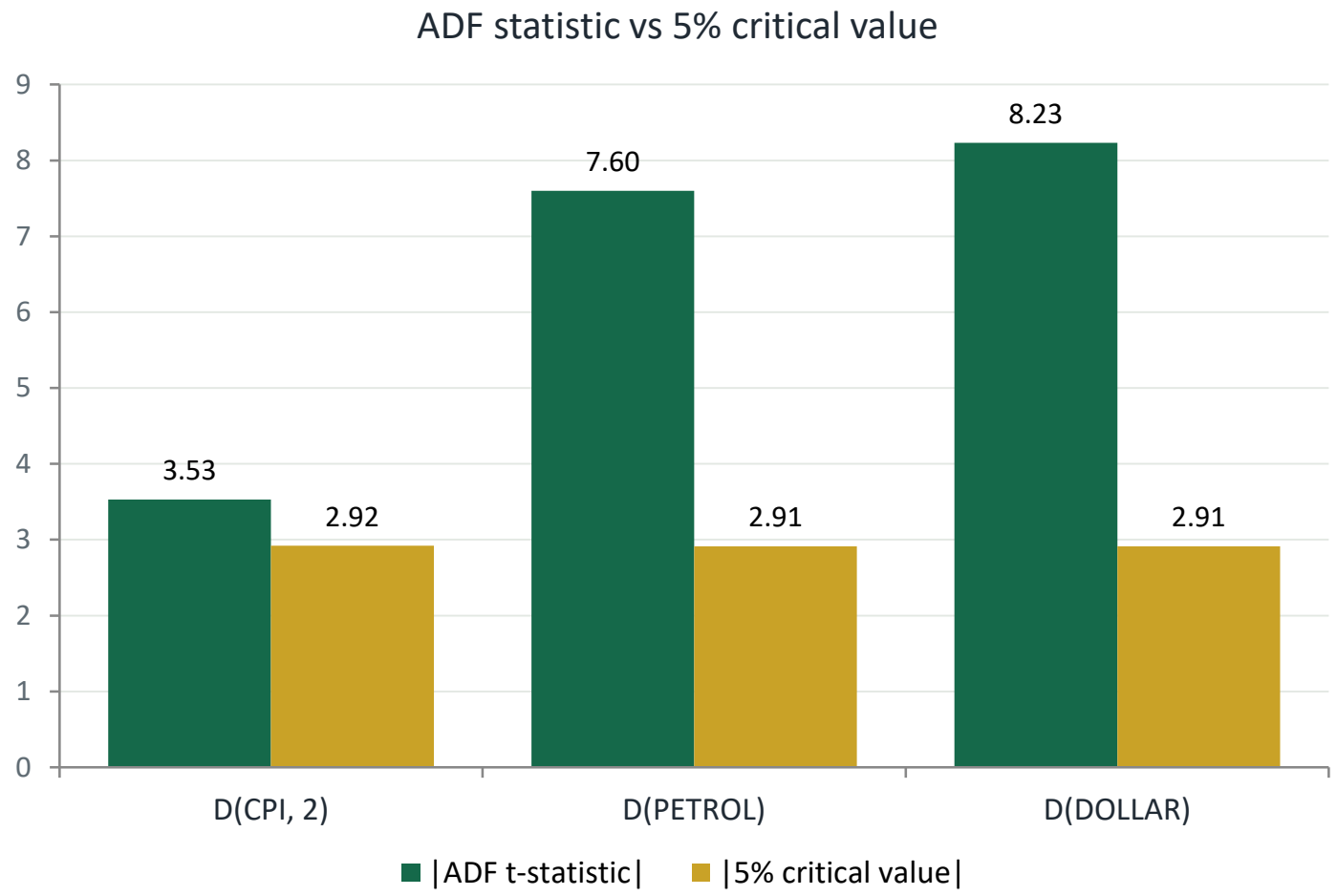
Date: 10/04/23 Time: 21:09
Sample (adjusted): 2018M10 2023M09
Included observations: 60 after adjustments



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	-0.116	-0.116	0.8529	0.356
		2	0.166	0.155	2.6235	0.269
		3	-0.171	-0.142	4.5231	0.210
		4	0.125	0.077	5.5627	0.234
		5	0.073	0.144	5.9258	0.314
		6	0.011	-0.028	5.9345	0.431
		7	0.161	0.173	7.7566	0.355
		8	0.050	0.112	7.9326	0.440
		9	0.004	-0.065	7.9337	0.541
		10	0.068	0.106	8.2825	0.601
		11	0.044	0.061	8.4266	0.675
		12	-0.001	-0.095	8.4266	0.751
		13	-0.034	-0.024	8.5167	0.808
		14	-0.088	-0.126	9.1439	0.822
		15	0.088	0.006	9.7806	0.833
		16	-0.085	-0.061	10.394	0.845
		17	0.001	-0.096	10.394	0.886
		18	0.039	0.092	10.531	0.913
		19	-0.032	-0.030	10.627	0.936
		20	0.015	-0.010	10.648	0.955
		21	-0.126	-0.022	12.165	0.935
		22	-0.103	-0.181	13.204	0.927
		23	-0.088	-0.072	13.976	0.928
		24	-0.046	0.002	14.195	0.942
		25	0.022	-0.047	14.246	0.957
		26	-0.019	0.032	14.283	0.969
		27	-0.106	-0.083	15.550	0.961
		28	-0.110	-0.122	16.951	0.950

D(DOLLAR)

Unit Root Test (Augmented Dickey-Fuller)



D(CPI, 2)

p-value = 0.0111

D(PETROL)

p-value = 0.0000

D(DOLLAR)

p-value = 0.0000

The unit root null is rejected for all three differenced series: they are stationary at the 5% level.

Granger Causality Test (Lag 2)

Null hypothesis	F-statistic	Prob.	Decision at 5%
D(PETROL) does not Granger cause D(CPI,2)	2.744	0.0737	Do not reject
D(DOLLAR) does not Granger cause D(CPI,2)	2.985	0.0593	Do not reject
D(CPI,2) does not Granger cause D(PETROL)	**4.476**	**0.0161**	**Reject H_o**
D(CPI,2) does not Granger cause D(DOLLAR)	1.268	0.2899	Do not reject
D(DOLLAR) does not Granger cause D(PETROL)	**4.919**	**0.0110**	**Reject H_o**
D(PETROL) does not Granger cause D(DOLLAR)	0.592	0.5570	Do not reject

Significant at 5%: CPI → Petrol and Dollar → Petrol. Dollar → CPI and Petrol → CPI are significant only at the 10% level.

VAR Optimal Lag Length Selection

VAR Lag Order Selection Criteria
Endogenous variables: D(CPI,2) D(PETROL) D(DOLLAR)
Exogenous variables: C
Date: 10/06/23 Time: 00:56
Sample: 2018M09 2023M09
Included observations: 54

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-553.3711	NA	178566.8	20.60634	20.71684*	20.64895
1	-537.6529	29.10778	139343.7	20.35751	20.79951	20.52797
2	-524.5684	22.77677	120211.2	20.20624	20.97973	20.50454
3	-509.4397	24.65408	96624.40	19.97925	21.08424	20.40540*
4	-499.1812	15.57775	93706.50	19.93264	21.36913	20.48663
5	-485.3441	19.47439*	80402.81*	19.75349*	21.52147	20.43533

* indicates lag order selected by the criterion
LR: sequential modified LR test statistic (each test at 5% level)
FPE: Final prediction error
AIC: Akaike information criterion
SC: Schwarz information criterion
HQ: Hannan-Quinn information criterion

Selection criteria

LR, FPE and AIC select lag 5. SC selects lag 0 and HQ selects lag 3. The estimated VAR uses 5 lags of each variable.

Estimated VAR Model

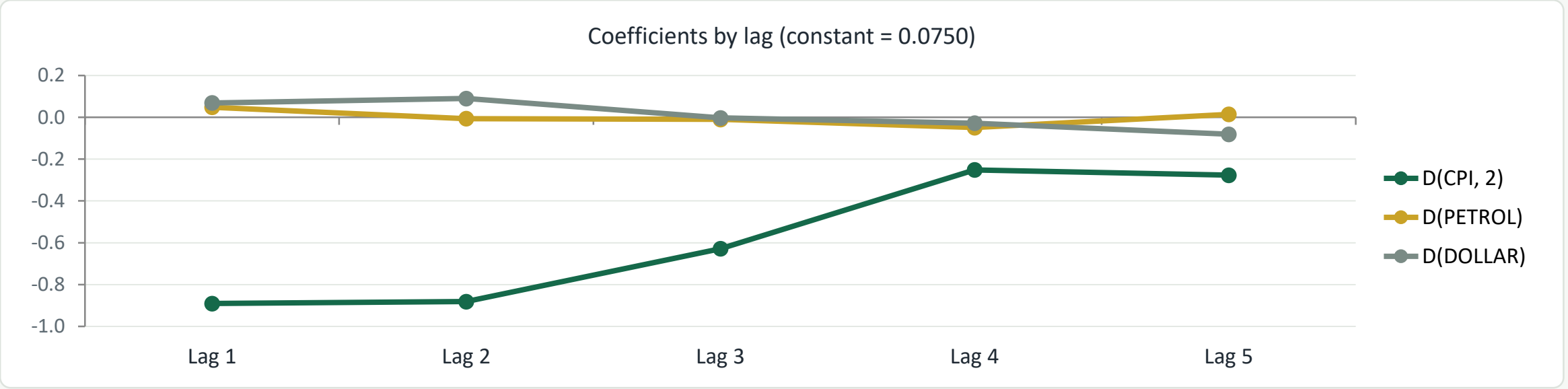
Vector Autoregression Estimates
Date: 10/06/23 Time: 00:47
Sample (adjusted): 2019M04 2023M09
Included observations: 54 after adjustments
Standard errors in ( ) & t-statistics in [ ]

	D(CPI,2)	D(PETROL)	D(DOLLAR)
D(CPI(-1),2)	-0.890696	1.896808	0.337542
	(0.15791)	(0.78567)	(0.45749)
	[-5.64061]	[2.41424]	[0.73781]
D(CPI(-2),2)	-0.881488	1.918131	-0.067249
	(0.21625)	(1.07596)	(0.62652)
	[-4.07626]	[1.78272]	[-0.10734]
D(CPI(-3),2)	-0.628679	0.094065	0.175378
	(0.23534)	(1.17095)	(0.68183)
	[-2.67134]	[0.08033]	[0.25722]
D(CPI(-4),2)	-0.251538	0.116878	-1.708199
	(0.21534)	(1.07143)	(0.62388)
	[-1.16810]	[0.10909]	[-2.73801]
D(CPI(-5),2)	-0.276842	-2.586290	-1.269565
	(0.19685)	(0.97946)	(0.57033)
	[-1.40633]	[-2.64053]	[-2.22603]
D(PETROL(-1))	0.048181	0.006624	-0.070612
	(0.03425)	(0.17041)	(0.09923)
	[1.40676]	[0.03887]	[-0.71161]
D(PETROL(-2))	-0.006397	0.027969	-0.058592
	(0.03434)	(0.17084)	(0.09948)
	[-0.18631]	[0.16371]	[-0.58899]
D(PETROL(-3))	-0.009969	-0.198729	0.162252
	(0.03397)	(0.16904)	(0.09843)
	[-0.29342]	[-1.17563]	[1.64839]
D(PETROL(-4))	-0.049128	0.132477	0.113529
	(0.03367)	(0.16755)	(0.09756)
	[-1.45889]	[0.79067]	[1.16365]
D(PETROL(-5))	0.014561	0.185305	-0.096459
	(0.03272)	(0.16279)	(0.09479)
	[0.44507]	[1.13834]	[-1.01763]
D(DOLLAR(-1))	0.068929	0.161177	0.043428
	(0.05801)	(0.28865)	(0.16808)
	[1.18815]	[0.55838]	[0.25838]
D(DOLLAR(-2))	0.090277	0.655610	0.411834
	(0.05863)	(0.29172)	(0.16987)
	[1.53974]	[2.24738]	[2.42445]
D(DOLLAR(-3))	-0.002459	-0.445418	-0.262322
	(0.05382)	(0.26778)	(0.15592)
	[-0.04569]	[-1.66338]	[-1.68236]
D(DOLLAR(-4))	-0.027875	-0.111517	-0.040118
	(0.05394)	(0.26837)	(0.15627)
	[-0.51680]	[-0.41554]	[-0.25673]
D(DOLLAR(-5))	-0.080670	0.234558	0.302936
	(0.05332)	(0.26528)	(0.15447)
	[-1.51300]	[0.88418]	[1.96110]
C	0.075045	2.171341	1.611579
	(0.44754)	(2.22674)	(1.29661)
	[0.16768]	[0.97512]	[1.24292]
R-squared	0.644778	0.543465	0.493193
Adj. R-squared	0.504559	0.363254	0.293138
Sum sq. resids	217.3018	5379.506	1823.980
S.E. equation	2.391332	11.89815	6.928164
F-statistic	4.598360	3.015709	2.465285
Log likelihood	-114.2149	-200.8596	-171.6571
Akaike AIC	4.822773	8.031838	6.950262
Schwarz SC	5.412102	8.621166	7.539590
Mean dependent	0.045185	4.409630	2.802481
S.D. dependent	3.397378	14.91063	8.240449

Determinant resid covariance (dof adj.)	36911.22
Determinant resid covariance	12862.57
Log likelihood	-485.3441
Akaike information criterion	19.75349
Schwarz criterion	21.52147
Number of coefficients	48

VAR Equation: D(CPI, 2) Coefficients

Regressor	Lag 1	Lag 2	Lag 3	Lag 4	Lag 5
D(CPI, 2)	-0.8907	-0.8815	-0.6287	-0.2515	-0.2768
D(PETROL)	0.0482	-0.0064	-0.0100	-0.0491	0.0146
D(DOLLAR)	0.0689	0.0903	-0.0025	-0.0279	-0.0807



None of the petrol or dollar coefficients has a t-statistic above 2 in absolute value; the own lags of CPI at lags 1 to 3 do.

Residual Serial Correlation LM Test

VAR Residual Serial Correlation LM Tests
Date: 10/06/23 Time: 00:58
Sample: 2018M09 2023M09
Included observations: 54

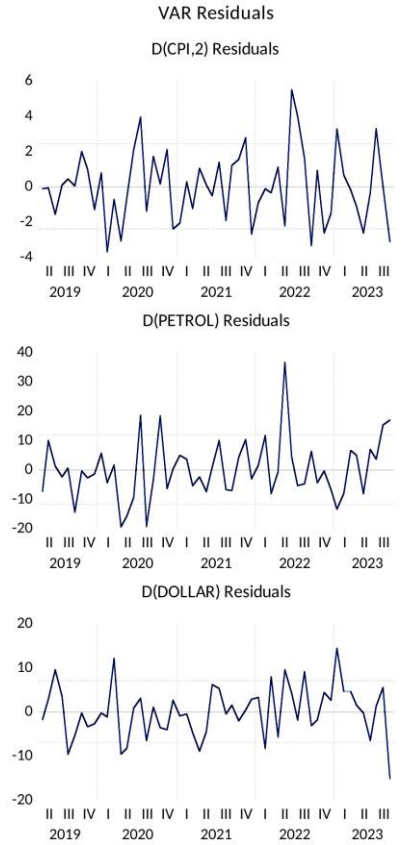
Null hypothesis: No serial correlation at lag h

Lag	LRE* stat	df	Prob.	Rao F-stat	df	Prob.
1	8.092213	9	0.5249	0.904543	(9, 80.5)	0.5255
2	13.69586	9	0.1336	1.584006	(9, 80.5)	0.1341
3	22.96450	9	0.0063	2.812331	(9, 80.5)	0.0064
4	12.67407	9	0.1779	1.456706	(9, 80.5)	0.1785
5	7.980056	9	0.5362	0.891402	(9, 80.5)	0.5368
6	6.106176	9	0.7292	0.674408	(9, 80.5)	0.7297

Null hypothesis: No serial correlation at lags 1 to h

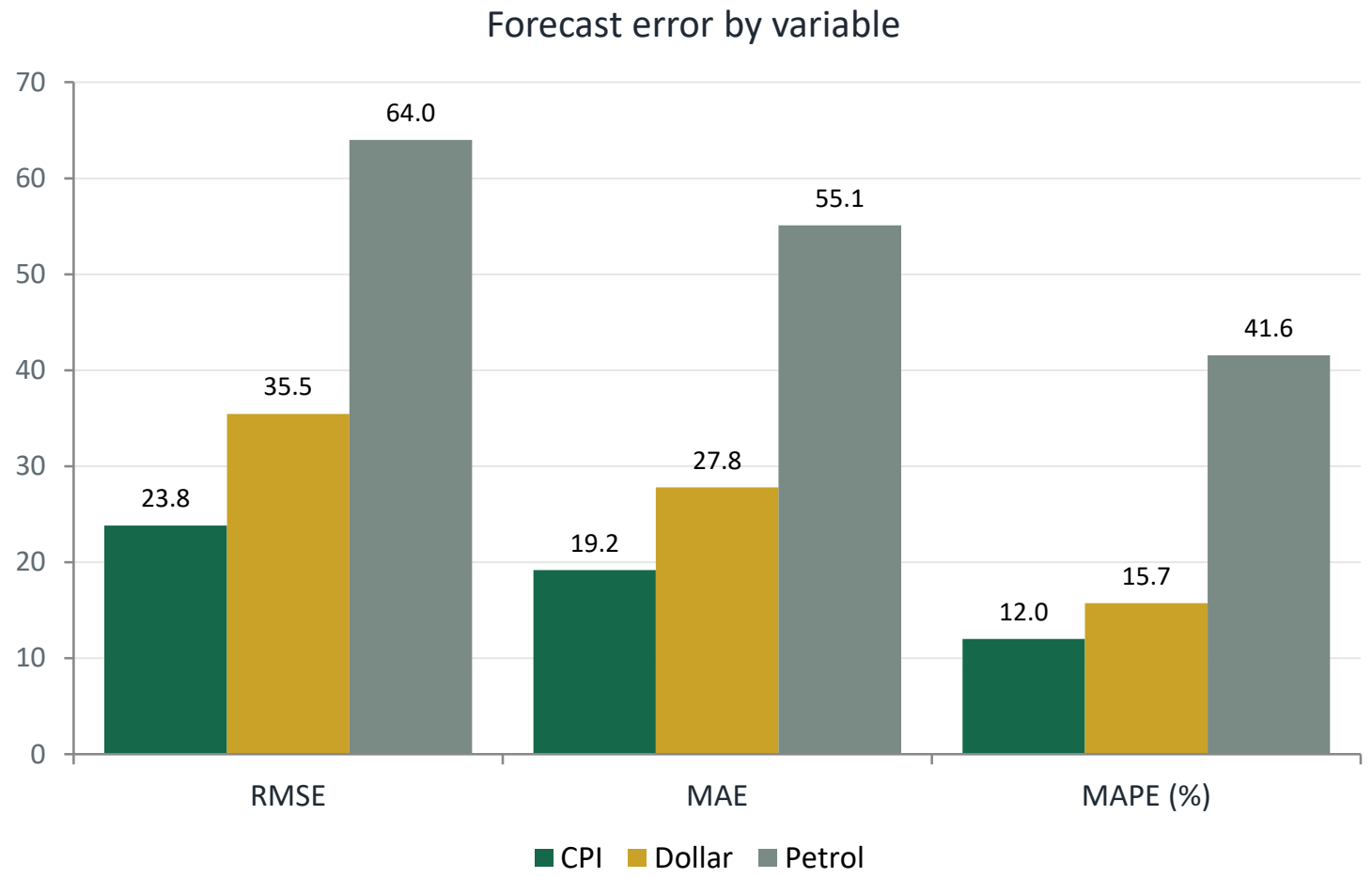
Lag	LRE* stat	df	Prob.	Rao F-stat	df	Prob.
1	8.092213	9	0.5249	0.904543	(9, 80.5)	0.5255
2	16.64723	18	0.5475	0.925674	(18, 85.3)	0.5507
3	39.87129	27	0.0527	1.597271	(27, 79.5)	0.0566
4	42.38125	36	0.2150	1.219973	(36, 71.6)	0.2346
5	48.33639	45	0.3397	1.080660	(45, 63.2)	0.3836
6	59.55091	54	0.2808	1.105538	(54, 54.4)	0.3564

*Edgeworth expansion corrected likelihood ratio statistic.



Serial correlation is detected only at lag 3 (p = 0.0063). The other lags show no significant autocorrelation in the residuals.

Forecast Evaluation



Theil inequality coefficient

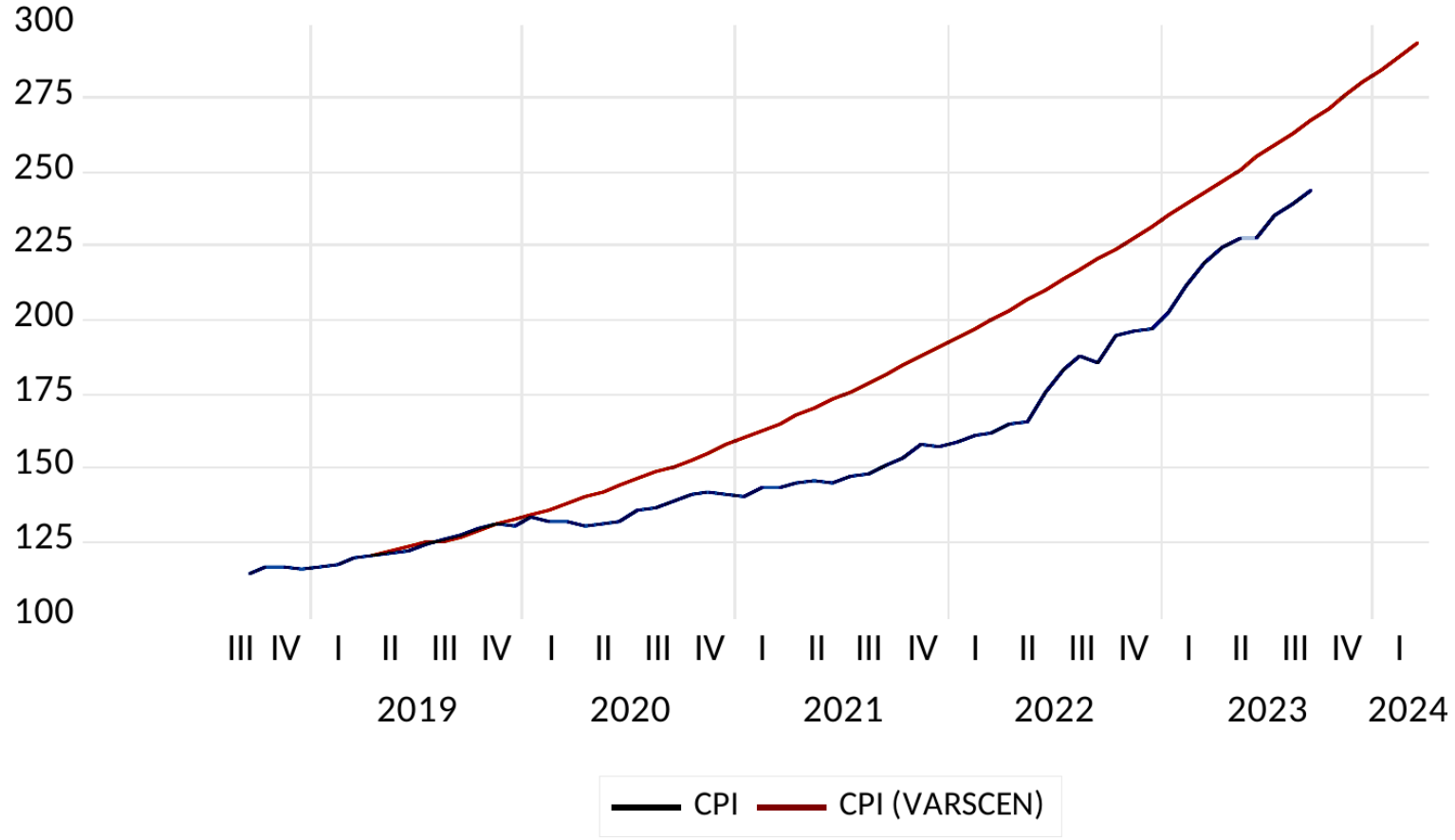
0.068 CPI

0.085 Dollar

0.163 Petrol

CPI has the lowest error on every measure.

CPI Forecast (VAR)



Blue: actual CPI. Orange: VAR forecast. The forecast path lies above actual CPI from 2021 onward, so treat it as indicative.

Conclusion

A VAR model was applied to Pakistan’s CPI, oil price and exchange rate (Sep 2018 to Sep 2023).

The forecast shows a continued rise in inflation.

Oil prices and the dollar exchange rate influence inflation: Granger causality toward CPI is significant at the 10% level.

The alternative hypothesis is accepted at the 10% significance level.

Recommendations

1. Monetary policy adjustment
2. Exchange rate management
3. Agricultural reforms
4. Energy sector reforms
5. Education and workforce development
6. Public awareness
7. Fiscal policy measures
8. Supply-side policies
9. Regulatory reforms
10. Financial inclusion
11. Inflation targeting
12. International collaboration

Coordinated efforts by the government, central bank and private sector, with continuous monitoring, are essential for stable growth and lower inflation.

References

- Aziz, M. I. (2009). Oil Price and Exchange Rate.
- Chen, S. and Chen, H. (2007). Oil prices and real exchange rate.
- Dickey, D. and Fuller, W. (1981). Likelihood ratio statistics for autoregressive time series with a unit root.
- Iqbal, F. and Razaq, A. (2018). Crude Oil Price-Exchange Rate Nexus in Pakistan. Financial Statistical Journal, 1(2).
- Kiani, A. (2008). Impact of High Oil Prices on Pakistan's Economic Growth. International Journal of Business and Social Science, 2(17), 209-216.
- Kutan, A. M. and Wyzan, M. L. (2005). Explaining the real exchange rate in Kazakhstan, 1996-2003.
- LeBlanc, M. and Chinn, M. D. (2004). Do High Oil Prices Presage Inflation?